

Exercise 1: Analyzing Marketing Campaign Performance

Problem Statement:

You are tasked with analyzing the performance of various marketing campaigns for a grocery app. The goal is to evaluate which campaign provides the best return on investment (ROI) and identify patterns in impressions, clicks, signups, and purchases.

Dataset:

The dataset contains information about five marketing campaigns. The columns include:

- **Campaign:** The name of the marketing campaign.
- **Spend:** The total cost of the campaign.
- **Impressions:** The number of people who saw the campaign.
- **Clicks:** The number of clicks the campaign received.
- **Signups:** The number of users who signed up via the campaign.
- **Purchases:** The number of purchases made by users acquired through the campaign.
- **Revenue:** The total revenue generated by the purchases.

Exercise Objectives:

1. **Calculate ROI** for each campaign using the formula: $ROI = \frac{Revenue - Spend}{Spend} \times 100$
 2. **Analyze Conversion Rates:**
 - **Click-through Rate (CTR):** $CTR = \frac{Clicks}{Impressions} \times 100$
 - **Signup Conversion Rate:** $Signup\ Conversion\ Rate = \frac{Signups}{Clicks} \times 100$
 - **Purchase Conversion Rate:** $Purchase\ Conversion\ Rate = \frac{Purchases}{Signups} \times 100$
 3. Identify which campaign performed best across different metrics (ROI, CTR, Signup Conversion, and Purchase Conversion)
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Exercise 2. Customer Acquisition Cost (CAC) Calculation

Problem Statement:

Analyze the impact of discounts on user acquisition and purchases for the grocery app. Your task is to determine how varying discount levels influence acquisition costs, purchase behavior, and overall return on investment (ROI).

Dataset:

The dataset contains information about campaigns offering different discount levels. The columns include:

- **Campaign:** The name of the marketing campaign.
- **Discount Offered (%):** The discount percentage provided in the campaign.
- **Spend:** The total cost of the campaign.
- **Impressions:** The number of people who saw the campaign.
- **Clicks:** The number of clicks the campaign received.
- **Signups:** The number of users who signed up via the campaign.
- **Purchases:** The number of purchases made by users acquired through the campaign.
- **Revenue:** The total revenue generated by the purchases.

Exercise Objectives:

1. Calculate **Cost per Signup** and **Cost per Purchase**:

- $\text{Cost per Signup} = \text{Spend} / \text{Signups}$
- $\text{Cost per Purchase} = \text{Spend} / \text{Purchases}$

2. Determine **Revenue per Purchase**: $\text{Revenue per Purchase} = \text{Revenue} / \text{Purchases}$

3. Calculate the **Discount ROI** using the formula: $\text{Discount ROI} = (\text{Revenue} - \text{Spend}) / \text{Spend} \times 100$

4. Identify patterns in acquisition efficiency and ROI based on discount level

Exercise 3. Funnel Analysis: From Impressions to Signups

Problem Statement:

Perform a funnel analysis to understand the user drop-off from impressions to clicks and then to signups for different marketing campaigns. Your goal is to calculate the conversion rates at each step of the funnel to identify the most and least effective campaigns.

Dataset:

The dataset tracks user flow through the following funnel steps:

1. **Impressions:** The number of people who saw the campaign.
2. **Clicks:** The number of people who clicked on the campaign.
3. **Signups:** The number of users who signed up after clicking.

The columns include data for five campaigns.

Exercise Objectives:

1. Calculate **Signup Conversion Rate** from Impressions to Signups for each campaign:
$$\text{Signup Conversion Rate} = \text{Signups} / \text{Impressions} \times 100$$
 2. Analyze which campaigns have the highest and lowest conversion rates.
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Exercise 4. Channel Attribution Analysis

Problem Statement:

Analyze advanced channel attribution to evaluate the performance of different acquisition sources (e.g., Email, Social Media, PPC, Affiliate, Influencer). The dataset includes multiple touchpoints for each user. Your task is to calculate the revenue contribution and user engagement for each source while factoring in the order of touchpoints (e.g., first-touch, second-touch, third-touch).

Dataset: The dataset contains:

- **User ID:** A unique identifier for each user.
- **Acquisition Source:** The channel through which the user interacted.
- **Revenue:** The revenue generated by the user.
- **Touchpoint Order:** The order in which the channel engaged the user (e.g., first, second, third).

Exercise Objectives:

1. **Overall Metrics:**
 - Total revenue for each acquisition source.
 - Number of users acquired through each source.
 - Average revenue per user (ARPU) for each source.
 2. **Touchpoint Analysis:**
 - Revenue contribution at each touchpoint order (first, second, third) for each source.
 - User count distribution by touchpoint order for each source.
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Exercise 5 Segmentation of New Users by Acquisition Source

Problem Statement:

Perform an advanced segmentation analysis of new users by their acquisition source. In addition to evaluating signups and purchases, assess their purchasing behavior, including average order value and time to first purchase. Use this data to determine the most valuable acquisition channels.

Dataset:

The dataset includes:

- **User ID:** A unique identifier for each user.
- **Acquisition Source:** The channel through which the user was acquired.
- **Signups:** Whether the user signed up (1 for signup, 0 for no signup).
- **Purchases:** Whether the user made a purchase (1 for purchase, 0 for no purchase).
- **Order Value:** The value of the user's purchase (if any).
- **Time to First Purchase (Days):** The number of days from signup to first purchase (if applicable).

Exercise Objectives:

1. Calculate the **Total Signups** and **Total Purchases** for each acquisition source.
 2. Compute the **Conversion Rate** for each acquisition source: $\text{Conversion Rate} = \frac{\text{Total Purchases}}{\text{Total Users}} \times 100$
 3. Determine the **Average Order Value (AOV)** for each acquisition source: $\text{AOV} = \frac{\text{Total Revenue}}{\text{Number of Purchases}}$
 4. Analyze the **Average Time to First Purchase (Days)** for users who made a purchase.
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Exercise 6: Trend Analysis in User Signups

Problem Statement:

Analyze trends in user signups for the grocery app over a 3-month period. The objective is to identify patterns and variations in signup behavior across different acquisition sources and timeframes.

Dataset:

The dataset contains:

- **Date:** The date of user activity.
- **Acquisition Source:** The channel through which the user was acquired (e.g., Email, Social Media, PPC, Affiliate, Influencer).
- **Signups:** The number of user signups recorded on that day.

Exercise Objectives:

1. Calculate the **Total Signups** for each acquisition source by month.
 2. Compute the **Average Daily Signups** for each source within each month.
 3. Identify trends in signup behavior, such as peak and slow periods for different channels.
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Exercise 7 AB Testing for Landing Pages

Problem Statement:

Conduct an A/B test analysis to compare the performance of two landing page versions (Group A and Group B). The goal is to identify which version leads to higher engagement and conversions for the grocery app.

Dataset:

The dataset includes:

- **User ID:** A unique identifier for each user.
- **Group:** The group the user was assigned to (A or B).
- **Clicks:** Whether the user clicked on the landing page (1 for yes, 0 for no).
- **Signups:** Whether the user signed up (1 for yes, 0 for no).
- **Purchases:** Whether the user made a purchase (1 for yes, 0 for no).
- **Time on Page (Seconds):** The amount of time the user spent on the landing page.

Exercise Objectives:

1. Calculate **Click-through Rate (CTR)** for each group: $CTR = \text{Total Clicks} / \text{Total Users} \times 100$
 2. Determine **Signup Rate** and **Purchase Rate** for each group: $\text{Signup Rate} = \text{Total Signups} / \text{Total Users} \times 100$
 3. $\text{Purchase Rate} = \text{Total Users} / \text{Total Purchases} \times 100$
 4. Analyze the **Average Time on Page** for each group.
 5. Identify the better-performing landing page based on engagement and conversion metrics.
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Exercise 8 Impact of Discounts on Acquisition

Problem Statement:

Conduct an A/B test analysis to compare the performance of two landing page versions (Group A and Group B). The goal is to identify which version leads to higher engagement and conversions for the grocery app.

Dataset:

The dataset includes:

- **User ID:** A unique identifier for each user.
- **Group:** The group the user was assigned to (A or B).
- **Clicks:** Whether the user clicked on the landing page (1 for yes, 0 for no).
- **Signups:** Whether the user signed up (1 for yes, 0 for no).
- **Purchases:** Whether the user made a purchase (1 for yes, 0 for no).
- **Time on Page (Seconds):** The amount of time the user spent on the landing page.

Exercise Objectives:

1. Calculate **Click-through Rate (CTR)** for each group: $CTR = \text{Total Clicks} / \text{Total Users} \times 100$
Determine **Signup Rate** and **Purchase Rate** for each group: $\text{Signup Rate} = \text{Total Signups} / \text{Total Users} \times 100$
 2. Analyze the **Average Time on Page** for each group.
 3. Identify the better-performing landing page based on engagement and conversion metrics.
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Exercise 9 Comprehensive Analysis of Marketing and Acquisition

In this assignment, you will apply all the concepts learned in the section, including awareness, acquisition, and early funnel metrics. Using the provided datasets, answer the following questions, perform analyses, and provide actionable insights.

Assignment Overview

You will work with a **comprehensive dataset** that consolidates key metrics from all exercises in Section 1. Your task is to:

1. Analyze marketing campaign performance.
2. Conduct funnel analysis.
3. Segment users by acquisition sources.
4. Assess trends in user signups.
5. Conduct advanced A/B testing.
6. Evaluate the impact of discounts on acquisition.

Datasets

Download the following consolidated dataset, which includes all the required data for your analyses:

Assignment Questions

Part 1: Marketing Campaign Performance Analysis

1. Calculate the **ROI** for each marketing campaign.
2. Analyze **click-through rates (CTR)** and **conversion rates** for campaigns:
 - Impressions → Clicks
 - Clicks → Signups
 - Signups → Purchases
3. Identify the top-performing campaign based on ROI, CTR, and conversion rates.

Part 2: Funnel Analysis

4. Perform a **funnel analysis**:
 - Calculate the drop-off rates at each stage (Impressions → Clicks → Signups → Purchases).
 - Visualize the funnel for the overall campaigns.
5. Provide actionable suggestions to reduce drop-offs at the weakest funnel stage.

Part 3: User Segmentation by Acquisition Source

6. Segment users by acquisition source:
 - Compute total signups, purchases, and conversion rates for each source.
 - Identify the source with the highest lifetime value (LTV).
7. Compare the performance of acquisition sources over time (e.g., month-over-month analysis).

Part 4: Trend Analysis of User Signups

8. Analyze **daily and monthly trends** in user signups:
 - Identify seasonal peaks or slow periods.
 - Compare trends across acquisition sources.
9. Suggest strategies for improving signup rates during slow periods.

Part 5: A/B Testing for Landing Pages

10. Conduct an **A/B test analysis**:
 - Compare CTR, Signup Rate, Purchase Rate, and Average Revenue per User (ARPU) for the two landing pages.
 - Identify the better-performing landing page.
11. Suggest improvements for the weaker landing page.

Part 6: Impact of Discounts

12. Evaluate the impact of discounts:
 - Analyze the relationship between discount percentage and user acquisition metrics (e.g., clicks, signups, purchases).
 - Compute ROI for each discount level.
 - Identify the optimal discount range for maximizing ROI.

Deliverables

1. **Analysis Report**:
 - Include clear visualizations for funnel analysis, trends, and A/B test results.
 - Provide data-driven recommendations for improving campaign performance and user acquisition.
2. **Code or Excel Work**:
 - Include all calculations in Excel or code scripts (if applicable).
 - Ensure clear documentation of steps and formulas used.